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Speciality

Balanced development through innovation and partnerships for sustainability and resilience.

Experience

Independent Mentor, Washington, DC (January 2022 to Present)

- Focus on a scalable low-impact development model in the Shastri River watershed in India
 - Distributed “Rurban” Services: Developing the “Water-Energy-Biomass” (WEB) initiatives to improve end-use efficiency, crop diversification and resource conservation for asset creation for the rural poor based on the demand side.
 - Watershed Management: Cultivating partnerships with local colleges and researchers, and industry associations toward climate change mitigation and adaptation measures for the coast, in the estuary, and at the watershed scale leveraging local aspirations.

Director and CEO, [ATREE](#), India, (January 2019 to December 2021)

Organization: Ashoka Trust for Research in Ecology and the Environment (ATREE) is a global non-profit organization focusing on applied research in sustainability science, and specializing in biodiversity conservation, water sustainability, and climate resilience. Staff: 250.

Mission: Generate rigorous interdisciplinary knowledge for achieving environmental conservation and sustainable development, to enable the use of this knowledge by policy makers and society, and to train the next generation of environmental leaders.

- *Partnerships and Initiatives*
 - Led the development and positioning of a national program with an international [Alliance to Reverse Ecosystem Service Threats \(AREST\)](#). Presented the findings at climate COP26 in Glasgow, coordinated with the National Mission on Biodiversity and Human Well-Being, and several ministries in India for demand-driven large-scale adaptation and sequestration.
 - Launched two new centers - Center for Social and Environmental Impact and Center for Policy Design – and mentored their leadership to leverage research findings into societal institutions such as the market, communities, cooperatives, etc., and into government.
 - Launched a [Masters degree program in Conservation Practice](#).

Independent Consultant, Washington, DC (2017-2018)

Strategic Advisor to the Chair of ATREE Board

Other initiatives:

- Cultivating strategy for new nanomaterials industry for restoring degraded lands
- Developing a data-driven tech startup for large impact of energy efficiency programs

Director, Priority Initiatives, [World Resources Institute](#), Washington, DC (2016-2017)

Organization: Not-for-profit 501(c)3 research organization in Washington, with affiliates in China, Brasil, Indonesia, Ethiopia, and Mexico, and projects in over 25 countries.

Mission: To move human society to live in ways that protect the Earth’s environment and its capacity to provide for the needs and aspirations of current and future generations.

- *Movement Building and Program Strategy*
 - Mobilized the [Global Restoration Council](#) (GRC) with research and communications from the Food, Forests, and Water programs toward impactful policies for scale up of the Bonn

Challenge. This movement aims to restore 150 million hectares of degraded land by 2020, and 350 million hectares by 2030 to forests and productive ecosystems.

- Conceptualized and delivered thought pieces as op-eds, blogs, presentations, etc., to reputed public figures on the GRC on restoration's linkages with employment, natural (water) infrastructure funding, and migration.
- Proposed alternative strategic entry points, e.g., green and cool roofs, for the organization to integrate an energy efficiency pillar to the existing global energy program focused on improving market share of solar photovoltaics in regulated utility markets.

CEO, World Resources Institute – India, Mumbai, India (2013-2016)

Reporting: Accountable to the Board of Trustees in India, with close operational alliance with the management of the WRI in Washington, DC. Staff: 65.

➤ Strategy and Leadership

- Developed a strategy, coordinated with the global WRI network, especially during the Paris climate change negotiations, for resource management in India.
- Cultivated bold, innovative entry points to lead diversification through interdisciplinary applications, e.g., Green Smart City-Regions, Distributed Energy for Utilities

➤ Innovation and Action

- Created themes, resources, and strategic partnerships (with unique organizations like [NABARD](#)) for diversification.
- Initiated a new cross-sectoral program by hiring high quality, key technical staff for restoration of degraded landscapes for climate change mitigation and adaptation.

President, [International Institute for Energy Conservation \(IIEC\)](#), Washington, DC (2000-2013)

Organization: Not-for-profit HQ in Washington, and offices in Bangkok, Thailand; Manila, Philippines; Pune and New Delhi, India. Staff: 25.

Offices: Also established and operated offices in Brazil, China, South Africa, and U.K.

Mission: Accelerate implementation of energy conservation programs on a massive scale for greenhouse gas reduction and sustainable development in developing countries

Reporting: Accountable to Board of Directors, including vision/strategy, fund raising, audits.

➤ Leadership and Results

- Motivated a diverse multi-national team of engineers, scientists, economists, architects to collaborate on several award winning impactful applications on the *demand side*.
- Worked with the Board to conceptualize go-to-market approaches to aggregate small transactions into pipelines to improve market penetration of energy efficient technologies.
- Negotiated a reverse merger, including legal and contractual separation.
- Developed and implemented outsourcing of management accounting functions.

➤ Innovation and Implementation

- Market Transformation: Guided award-winning programs for growing end-user markets for energy efficient equipment in China, India, Philippines, and South Africa. Developed global partnerships for [standards](#) and [labeling](#) policies, [harmonization practices](#) and [competitive mechanisms](#) for appliances in South Asia and APEC.
- Energy Efficiency Finance: Guided the development of assessment, monitoring and verification systems for achieving specific greenhouse gas emission reduction targets through the implementation of a 100 million Euro credit line from KfW for a portfolio of investments in SMEs in India. Also worked with states in India to launch the first public benefit charge based clean energy fund for energy efficiency and renewable energy.
- Demand Side Management (DSM): Cultivated a relationship with private distribution utilities in several states in India, including Reliance Energy Limited (REL), Tata Power Company (TPC), and Bangalore Electricity Supply Company (BESCOM), to develop DSM

programs. Integrated renewable energy systems (solar thermal) at the end user level as a DSM measure. Developed innovative “feeder-based” bidding and standard offer based demand acquisition systems for the integration of new technologies in the grid with utility policy and regulatory commissions in Maharashtra, Karnataka, and Puducherry states to finance and to enable distribution utilities and Super ESCOs ([EESL](#)) to aggregate, market and purchase demand from local energy service entrepreneurs. Conducted utility DSM pilots (World Bank, UNDP, USAID, UK-FCO) in Vietnam, India, Laos, South Africa, and Sri Lanka to improve end use energy efficiency.

- **Urban Programs and Sustainable Habitat:** Created the [Mumbai Energy Alliance](#) to bring together city stakeholders to develop policies and processes to finance measures energy efficiency measures, such as end-use and municipal water pumping to address energy and power deficits, including research on alternative climate-friendly construction materials and techniques, and an innovative property tax incentive for green roofs on new commercial buildings in India. Developed brands in India for [construction of buildings for residential energy and water conservation](#), and for [low- and middle-income urban and peri-urban communities](#) in South Africa, and mandatory energy efficiency commercial building codes in Thailand, Vietnam, and India.

President, ENSYS, INC., Vienna, VA, USA (1991 - 2000)

Founded and ran a boutique entrepreneurial high-tech for-profit consultancy specializing in niche applications of state-of-the-art artificial intelligence (AI) techniques, policy methods, and offset counter-trading for renewable energy, water quality and environmental problem solving. Staff: 5

- Conducted a special assignment for Lockheed Martin Environmental Systems for the development and delivery of international industrial offset programs for offset program planning and obligations in several countries including Philippines, Canada, Norway, Netherlands and India. For example:
 - Managed techno-economic feasibility studies for biomass cogeneration and reuse of agricultural wastes in India (40 MW) and Philippines (15 MW) using Canadian, Dutch and Norwegian technologies for a multi-national consortium. Conducted a comprehensive evaluation of biomass for cogeneration, optimal sizing for captive steam/power, use of alternative energy efficient processes, market R&D to manufacture new materials, analysis of grid interconnection of renewable energy, proforma development for financing.
- International Consulting:
 - Developed a portfolio risk knowledge management system using *artificial intelligence (AI)* techniques for managing the carbon dioxide emission projects portfolio for Enron.
 - Created an expert (AI) system for modeling tradable coal fired power plant emissions for the Asian Development Bank, including estimation of point source sulfur dioxide emissions.
 - Developed novel geographic information system (GIS) based management of nutrient offset trading for point/non-point source pollution control in Tar-Pamlico basin in North Carolina.
- Implemented Small Business Innovative Research (SBIR) award programs to utilize GIS/AI tools to use synoptic climatologies in data sparse regions for US Air Force relief missions.

Project Director, Roy F. Weston, Inc., Washington, DC, USA (1986 - 1991)

Led a multi-disciplined applied R&D department to develop intelligent products for solving geo-environmental problems for U.S. Environmental Protection Agency and other clients using AI techniques, especially in hazardous waste management, and in geo-hydrologic systems. Staff: 5. Main products included:

- Computer-Aided Remedial Technologies Selector (CARTS) – an object-oriented AI system using non-monotonic reasoning implemented in Lisp and KEE™.

- Expert systems to (a) assist in operation of dam gates in the Sierra Nevada for water and power, and advise emergency personnel on chemical risk - Health & Safety Program (HASP).

Senior Geotechnical Engineer, Woodward-Clyde Consultants, Eastern Group, USA (1981 - 1984)

- Conducted geotechnical characterization, modeling, and design with field and lab interactions in USA. Piloted novel construction techniques, e.g., geo-grids for pavement reinforcement.
- Analyzed and managed geotechnical engineering applications including power plant fly ash disposal, ground-water modeling, tunneling in soil and rock strata, and construction of offshore and terrestrial foundations.

Geohydrology Engineer Trainee, Dubon Project Engineering Pvt. Ltd., Bombay, India (1978 - 1979)

Worked on water resource systems engineering for irrigation, drainage, and crop development projects in East Africa, and low-cost local biomass based construction materials in India.

Education

Doctor of Philosophy in Public Policy (1998); Major: Energy, Institutional Economics

The Institute of Public Policy, George Mason University, Fairfax, VA, USA

Thesis: A Common Property Approach for Retail Energy Institutions for Distributed Generation

Master of Science in Civil & Environmental Engineering (1986); Major: Artificial Intelligence

The Robotics Institute, Carnegie-Mellon University, Pittsburgh, PA, USA

Thesis: An Expert System for Mining Machine Condition Monitoring

Master of Science in Civil Engineering (1980); Major: Geotechnical Engineering

Fritz Engineering Laboratory, Lehigh University, Bethlehem, PA, USA

Thesis: Geotechnical Properties of Shanghai Soft Silt

Bachelor of Engineering (Honors) in Civil Engineering (1979); Major: Water Resource Systems

Birla Institute of Technology and Science, Pilani, Rajasthan, India

Thesis: Water Resource Development Methodology for West African Watersheds

Affiliations

Adjunct Professor of Public Policy, George Mason University, Fairfax, VA, USA. 2009

Registered Professional Engineer in State of Virginia, USA. Member: ACM, WEF, IEEE, APPA.

Skills, Publications, Patents

Languages: Fluent (written and verbal) in English, Hindi, Marathi.

Computers: Skilled in software use, including open source and artificial intelligence tools.

Publications: Several published papers; numerous public presentations and awards.

U.S. Patent 6,529,890 B1: "Method of representing synoptic climatology information in a class-object-attribute hierarchy and an expert system for obtaining synoptic climatology information"

Hobbies and Extracurricular

Swimming, tennis, walking and reading.